

A Technical Introduction to Precision and recall

$$\text{Precision} = \frac{TP}{TP+FP}, \quad \text{Recall} = \frac{TP}{TP+FN}$$

Section 1

Predicted positive condition rate = $\frac{TP}{TP+FP}$ $\frac{TP}{TP+FP+TN+FN}$ According to Saito and Rehmsmeier, precision-recall plots are more informative than ROC plots when evaluating binary classifiers on imbalanced data. In such scenarios, ROC plots may be visually deceptive with respect to conclusions about the reliability of classification performance. Different from the above approaches, if an imbalance scaling is applied directly by weighting the confusion matrix elements, the standard metrics definitions still apply even in the case of imbalanced datasets. The weighting procedure relates the confusion matrix elements to the support set of each considered class.

Section 2

This measure is approximately the average of the two when they are close, and is more generally the harmonic mean, which, for the case of two numbers, coincides with the square of the geometric mean divided by the arithmetic mean. There are several reasons that the F-score can be criticized, in particular circumstances, due to its bias as an evaluation metric. This is also known as the F_1 measure, because recall and precision are evenly weighted. It is a special case of the general F_β measure (for non-negative real values of β):

Section 3

For the previous example (95 negative and 5 positive samples), classifying all as negative gives 0.5 balanced accuracy score (the maximum bACC score is one), which is equivalent to the expected value of a random guess in a balanced data set. Balanced accuracy can serve as an overall performance metric for a model, whether or not the true labels are imbalanced in the data, assuming the cost of FN is the same as FP. The TPR and FPR are a property of a given classifier operating at a specific threshold. However, the overall number of TPs, FPs etc depend on the class imbalance in the data via the class ratio $r = P/N$. As the recall (or TPR) depends only on positive cases, it is not affected by r , but the precision is. We have that

Section 4

$$F = 2 \cdot \text{precision} \cdot \text{recall} / (\text{precision} + \text{recall})$$

Section 5

In pattern recognition, information retrieval, object detection and classification (machine learning), precision and recall are performance metrics that apply to data retrieved from a collection, corpus or sample space. Precision (also called positive predictive value) is the fraction of relevant instances among the retrieved instances. Written as a formula:

Section 6

In a classification task, the precision for a class is the number of true positives (i.e. the number of items correctly labelled as belonging to the positive class) divided by the total number of elements labelled as belonging to the positive class (i.e. the sum of true positives and false positives, which are items incorrectly labelled as belonging to the class). Recall in this context is defined as the number of true positives divided by the total number of elements that actually belong to the positive class (i.e. the sum of true positives and false negatives, which are items which were not labelled as belonging to the positive class but should have been). Precision and recall are not particularly useful metrics when used in isolation. For instance, it is possible to have perfect recall by simply retrieving every single item. Likewise, it is possible to achieve perfect precision by selecting only a very small number of extremely likely items. In a classification task, a precision score of 1.0 for a class C means that every item labelled as belonging to class C does indeed belong to class C (but says nothing about the number of items from class C that were not labelled correctly) whereas a recall of 1.0 means that every item from class C was labelled as belonging to class C (but says nothing about how many items from other classes were incorrectly also labelled as belonging to class C). Often, there is an inverse relationship between precision and recall, where it is possible to increase one at the cost of reducing the other, but context may dictate if one is more valued in a given situation: A smoke detector is generally designed to commit many Type I errors (to alert in many situations when there is no danger), because the cost of a Type II error (failing to sound an alarm during a major fire) is prohibitively high. As such, smoke detectors are designed with recall in mind (to catch all real danger), even while giving little weight to the losses in precision (and making many false alarms). In the other direction, Blackstone's ratio, "It is better that ten guilty persons escape than that one innocent suffer," emphasizes the costs of a Type I error (convicting an innocent person). As such, the criminal justice system is geared toward precision (not convicting innocents), even at the cost of losses in recall (letting more guilty people go free). A brain surgeon removing a cancerous tumor from a patient's brain

illustrates the tradeoffs as well: The surgeon needs to remove all of the tumor cells since any remaining cancer cells will regenerate the tumor. Conversely, the surgeon must not remove healthy brain cells since that would leave the patient with impaired brain function. The surgeon may be more liberal in the area of the brain they remove to ensure they have extracted all the cancer cells. This decision increases recall but reduces precision. On the other hand, the surgeon may be more conservative in the brain cells they remove to ensure they extract only cancer cells. This decision increases precision but reduces recall. That is to say, greater recall increases the chances of removing healthy cells (negative outcome) and increases the chances of removing all cancer cells (positive outcome). Greater precision decreases the chances of removing healthy cells (positive outcome) but also decreases the chances of removing all cancer cells (negative outcome). Usually, precision and recall scores are not discussed in isolation. A precision-recall curve plots precision as a function of recall; usually precision will decrease as the recall increases. Alternatively, values for one measure can be compared for a fixed level at the other measure (e.g. precision at a recall level of 0.75) or both are combined into a single measure. Examples of measures that are a combination of precision and recall are the F-measure (the weighted harmonic mean of precision and recall), or the Matthews correlation coefficient, which is a geometric mean of the chance-corrected variants: the regression coefficients Informedness (DeltaP') and Markedness (DeltaP). Accuracy is a weighted arithmetic mean of Precision and Inverse Precision (weighted by Bias) as well as a weighted arithmetic mean of Recall and Inverse Recall (weighted by Prevalence). Inverse Precision and Inverse Recall are simply the Precision and Recall of the inverse problem where positive and negative labels are exchanged (for both real classes and prediction labels). True Positive Rate and False Positive Rate, or equivalently Recall and $1 - \text{Inverse Recall}$, are frequently plotted against each other as ROC curves and provide a principled mechanism to explore operating point tradeoffs. Outside of Information Retrieval, the application of Recall, Precision and F-measure are argued to be flawed as they ignore the true negative cell of the contingency table, and they are easily manipulated by biasing the predictions. The first problem is 'solved' by using Accuracy and the second problem is 'solved' by discounting the chance component and renormalizing to Cohen's kappa, but this no longer affords the opportunity to explore tradeoffs graphically. However, Informedness and Markedness are Kappa-like renormalizations of Recall and Precision, and their geometric mean Matthews correlation coefficient thus acts like a debiased F-measure.